

Definition:

A relation $\sim$ on a set S is called an *equivalence relation* if the following axioms hold:

1. (Reflexivity) $a \sim a$
2. (Symmetry) $a \sim b$ implies $b \sim a$
3. (Transitivity) $a \sim b$ and $b \sim c$ implies $a \sim c$

1. One of the most important equivalence relations we'll deal with this semester is $\sim_n$, defined by $a \sim_n b$ if $n \mid (b - a)$. For this equivalence relation...

(a) Re-phrase the reflexivity statement into a statement about a , b and n , then prove it.

(b) Re-phrase the symmetry statement into a statement about a and n , then prove it.

(c) Re-phrase the transitivity statement into a statement about a , b , c and n , then prove it.

Consider the following relations. Either prove that the given relation is, in fact, an equivalence relation on the given set, or decide which of the three axioms fail to hold. If it is an equivalence relation, give a complete (non-repeating) list of the equivalence classes.

2. For $x, y \in \mathbb{R}$, we define $x \sim_d y$ if and only if $|x - y| < 5$.

3. For $x, y \in \mathbb{R}$, we define $x \sim_f y$ if and only if $\lfloor x \rfloor - \lfloor y \rfloor = 0$.

4. For $(a, b) \in \mathbb{R}^2$ and $(c, d) \in \mathbb{R}^2$, we define $(a, b) \sim_o (c, d)$ if and only if $a^2 + b^2 = c^2 + d^2$.

Equivalence Relations – Solutions

1. We re-formulate the three conditions as a series of claims, one in each part:

(a) **Claim:** $n \mid (a - a)$

Proof. This is simply saying that $n \mid 0$. Remember we defined the statement “ $n \mid m$ ” to mean that there is an integer k with $m = nk$. Thus $n \mid 0$, as we can take the integer to be zero: $0 = n \cdot 0$. $\square$

(b) **Claim:** $n \mid (b - a)$ implies $n \mid (a - b)$.

Proof. Here we assume that there is an integer k with $b - a = nk$ and we wish to prove that there is an integer ℓ with $a - b = n\ell$. And of course there must be: simply take $\ell = -k$. $\square$

(c) **Claim:** $n \mid (b - a)$ and $n \mid (c - b)$ implies $n \mid (c - a)$.

Proof. Here we assume that there are integers k with $b - a = nk$ and ℓ with $c - b = n\ell$, and we wish to prove that there is an integer m with $c - a = nm$. Since $c - a = (c - b) + (b - a)$, we add and find that $c - a = nk + n\ell = n(k + \ell)$, so we can take $m = k + \ell$. $\square$

2. This is *not* an equivalence relation. The given relation is reflexive (it is certainly true that $|x - x| < 5$) and symmetric (if $|x - y| < 5$, then $|y - x| < 5$ as well), but the transitive property fails. As an example, take $x = 0$, $y = 3$ and $z = 6$. Then $x \sim_d y$ (since $|x - y| = 3$) and $y \sim_d z$ (since $|y - z| < 3$), but $x \not\sim_d z$ (since $|x - z| = 6 \geq 5$).

The intuition is that $x \sim_d y$ if x and y are “close enough” to each other (in this case less than a distance of 5). So this explains why transitivity fails: simply because x is “close” to y and y is “close” to z , that doesn’t mean that x must be “close” to z . Finally, it also explains the notation: the subscript “d” is for *distance*.

3. Just to be clear, the function “ $\lfloor x \rfloor$ ” is the *floor* function: $\lfloor x \rfloor$ gives the integer part of x (that is, the greatest integer $\leq x$). (This explains the “f” subscript in $x \sim_f y$.) Our relation is $x \sim_f y$ if $\lfloor x \rfloor = \lfloor y \rfloor$. The three conditions for $\sim_f$ to be an equivalence relation are:

1. (Reflexivity) $\lfloor x \rfloor = \lfloor x \rfloor$,
2. (Symmetry) $\lfloor x \rfloor = \lfloor y \rfloor$ implies $\lfloor y \rfloor = \lfloor x \rfloor$, and
3. (Transitivity) $\lfloor x \rfloor = \lfloor y \rfloor$ and $\lfloor y \rfloor = \lfloor z \rfloor$ implies $\lfloor x \rfloor = \lfloor z \rfloor$.

These are all clear from the statements.

The equivalence classes here can all be represented by integers. That is, every real number x is equivalent to an integer: If $k = \lfloor x \rfloor$, then $x \sim_f k$. Thus the set of equivalence classes is $\mathbb{Z}$.

4. The three conditions for $\sim_o$ to be an equivalence relation are:

1. (Reflexivity) “ $(a, b) \sim_o (a, b)$ ” means $a^2 + b^2 = a^2 + b^2$,
2. (Symmetry) “ $(a, b) \sim_o (c, d)$ implies $(c, d) \sim_o (a, b)$ ” means “ $a^2 + b^2 = c^2 + d^2$ implies $c^2 + d^2 = a^2 + b^2$, and

3. (Transitivity) “ $(a, b) \sim_{\circ} (c, d)$ and $(c, d) \sim_{\circ} (e, f)$ implies $(a, b) \sim_{\circ} (e, f)$ ” means “ $a^2 + b^2 = c^2 + d^2$ and $c^2 + d^2 = e^2 + f^2$ implies $a^2 + b^2 = e^2 + f^2$ ”.

These are all clear from the statements.

Here two points (a, b) and (c, d) are in the same equivalence class if they are the same distance from the origin (that is, if $\sqrt{a^2 + b^2} = \sqrt{c^2 + d^2}$). Thus the equivalence classes can be represented by concentric circles centered at the origin. One way to choose a single representative from each equivalence class is to pick a ray from the origin that intersects each circle exactly once. Here's a picture with two sample rays:

